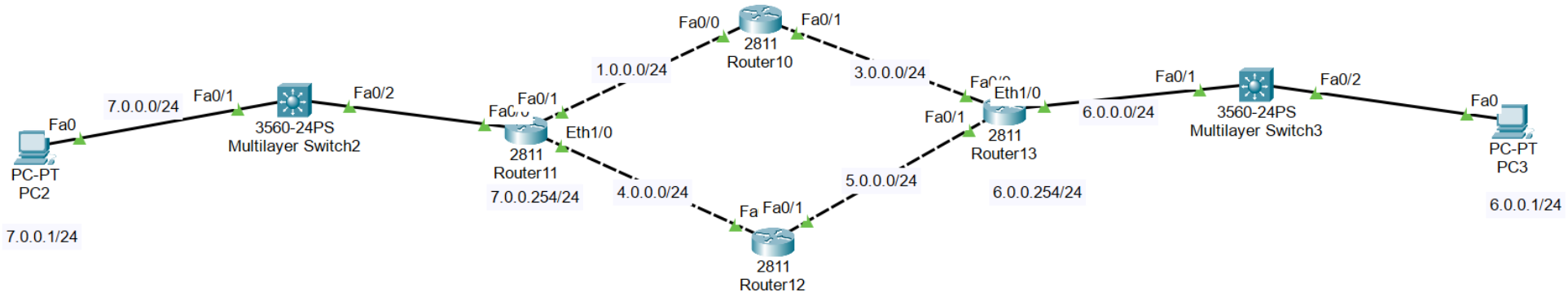
RIPv2 network lab

Topology, router configuration and end-to-end validation

Two LANs. Two transit branches. One observed forwarding path.

This lab connects PC2 on 7.0.0.0/24 to PC3 on 6.0.0.0/24 through a four-router diamond. The supplied screenshots capture Router11 configuration, a successful traceroute and a ping with an initial timeout.



Observed result

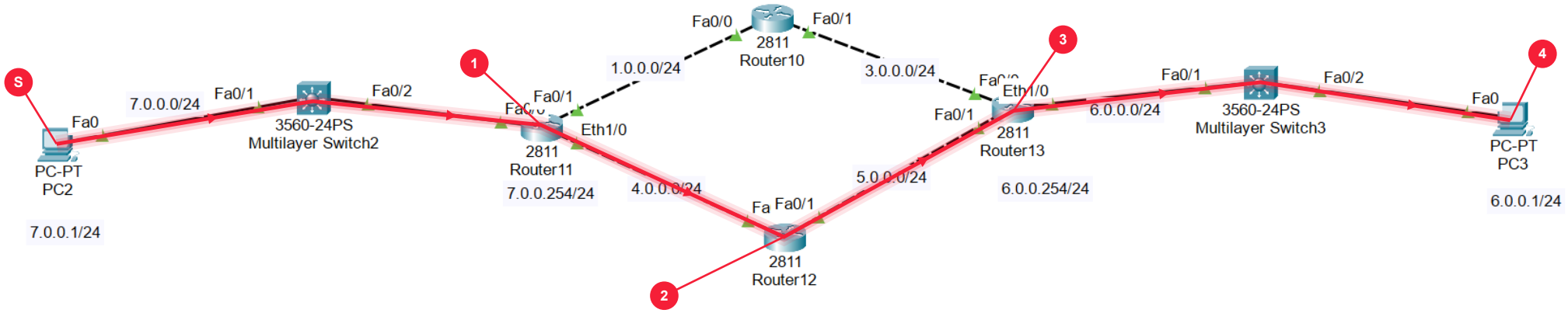
PC2 reaches PC3 through Router11, Router12 and Router13. The traceroute completes at hop 4.

Evidence boundary

Five screenshots supplied. No .pkt file, complete router configurations or routing-table captures were available for direct verification.

The traced route, in bright red

Original topology preserved beneath a vector path overlay. Arrows indicate the observed source-to-destination direction.



Marker	Device / response	Role in the captured trace
S	PC2 - 7.0.0.1	Source; not a traceroute hop
1	Router11 - 7.0.0.254	Default gateway / first router
2	Router12 - 4.0.0.2	Lower branch of the diamond
3	Router13 - 5.0.0.2	Router connected to the destination LAN
4	PC3 - 6.0.0.1	Destination; trace completes

Addressing and forwarding logic

All six subnet labels in the original diagram use /24 (255.255.255.0).

Subnet	Attached devices	Evidence / purpose
7.0.0.0/24	PC2 - Switch2 - Router11	PC2 .1; Router11 .254
1.0.0.0/24	Router11 - Router10	Router11 .1; upper-left transit
3.0.0.0/24	Router10 - Router13	Upper-right transit
4.0.0.0/24	Router11 - Router12	Router11 .1; Router12 .2 in trace
5.0.0.0/24	Router12 - Router13	Router13 .2 in trace
6.0.0.0/24	Router13 - Switch3 - PC3	Router13 .254; PC3 .1

How the packet crosses the topology

PC2 sends traffic for the remote 6.0.0.0/24 LAN to its expected gateway, 7.0.0.254. Router11 forwards along the lower branch to Router12, which forwards to Router13. Router13 delivers the packet onto the destination LAN. The two 3560 switches are shown as LAN access devices; Layer 2 forwarding does not add a traceroute hop.

The upper branch through Router10 is an alternate physical path. With matching RIP configuration and link metrics, both branches can have equal hop cost. The captured trace proves the lower branch was used by these probes; it does not establish that the upper branch is unavailable or that load sharing is active.

What the configuration screenshots show

The following extracts are captured evidence from Router11, inferred from the matching interface addresses.

```
!
!
!
interface FastEthernet0/0
 ip address 7.0.0.254 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 1.0.0.1 255.255.255.0
 duplex auto
 speed auto
!
interface Ethernet1/0
 ip address 4.0.0.1 255.255.255.0
 duplex auto
 speed auto
!
interface Ethernet1/1
```

```
!
router rip
 version 2
 network 1.0.0.0
 network 4.0.0.0
 network 7.0.0.0
 no auto-summary
!
```

RIP process

The extract enables RIP version 2, includes connected major networks 1.0.0.0, 4.0.0.0 and 7.0.0.0, and disables automatic summarization. [1]

Confirmed interface addresses

FastEthernet0/0: 7.0.0.254/24 FastEthernet0/1: 1.0.0.1/24
Ethernet1/0: 4.0.0.1/24. The cropped extract does not show interface status or a complete router configuration.

Configuration snippets for the lab

Router11 addresses and RIP settings are evidenced; activation commands and peer-router blocks below are suggested reproduction steps.

Router11 - interfaces and RIP

```
enable
configure terminal
interface FastEthernet0/0
 ip address 7.0.0.254 255.255.255.0
 no shutdown
interface FastEthernet0/1
 ip address 1.0.0.1 255.255.255.0
 no shutdown
interface Ethernet1/0
 ip address 4.0.0.1 255.255.255.0
 no shutdown
exit
router rip
 version 2
 network 1.0.0.0
 network 4.0.0.0
 network 7.0.0.0
 no auto-summary
end
copy running-config startup-config
```

Peer routers - suggested RIP blocks

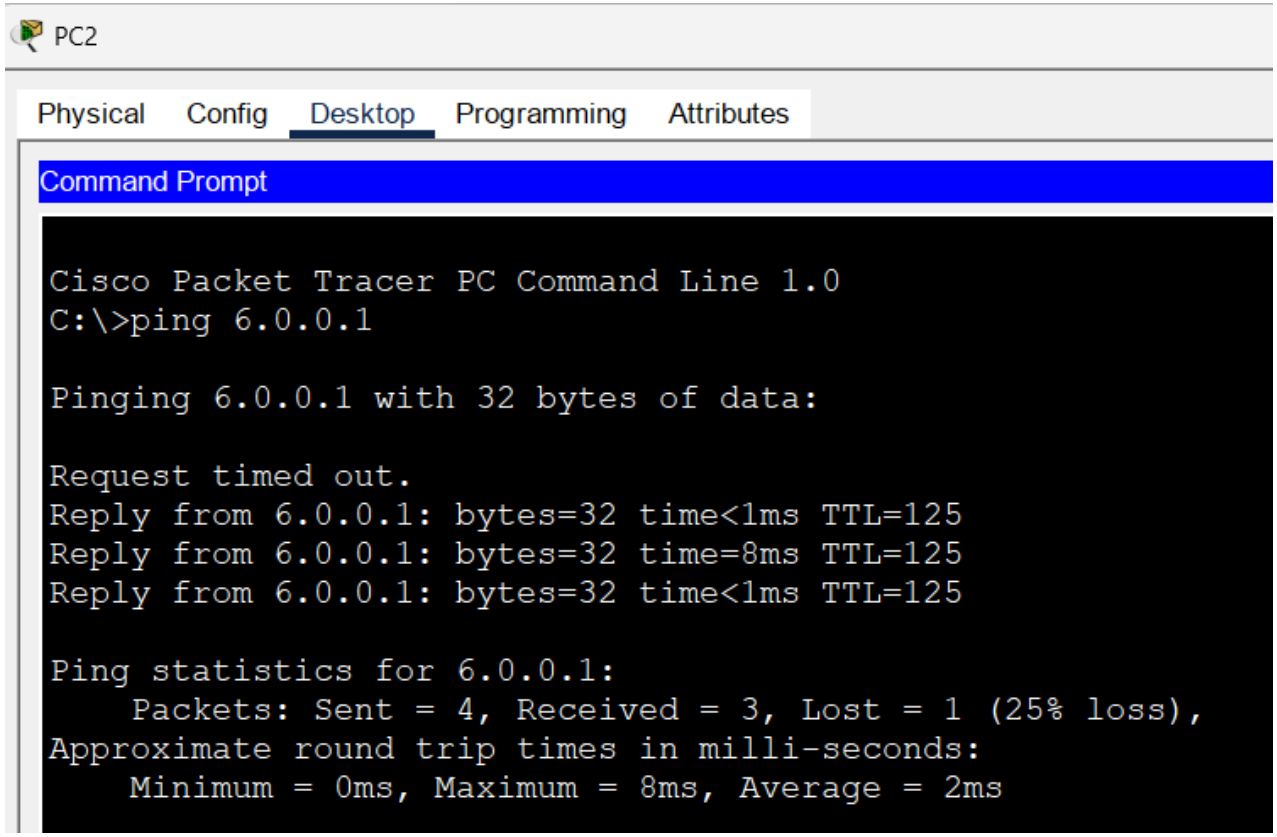
```
! Router10
router rip
 version 2
 network 1.0.0.0
 network 3.0.0.0
 no auto-summary
! Router12
router rip
 version 2
 network 4.0.0.0
 network 5.0.0.0
 no auto-summary
! Router13
router rip
 version 2
 network 3.0.0.0
 network 5.0.0.0
 network 6.0.0.0
 no auto-summary
```

Apply each peer block separately in global configuration mode on the named router, then exit and save. Configure and enable its connected interfaces first; the screenshots do not disclose every peer address or interface mapping.

PC settings: PC2 7.0.0.1/24, gateway 7.0.0.254; PC3 6.0.0.1/24, gateway 6.0.0.254. Gateway values are the intended LAN router addresses; PC gateway dialogs were not supplied.

PC2 reaches 6.0.0.1

Original command-prompt screenshot, with the initial timeout retained.



Captured test

Command: ping 6.0.0.1

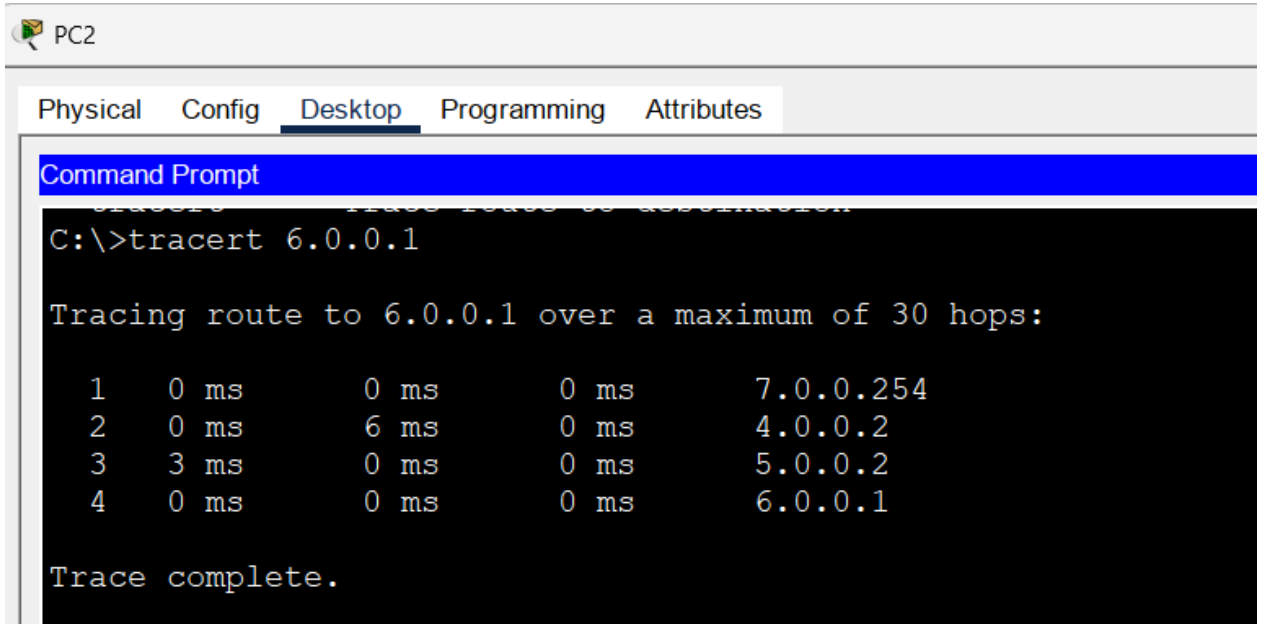
Payload: 32 bytes
Sent / received: 4 / 3
Lost: 1 (25%)
RTT min / max / avg:
0 / 8 / 2 ms
Reply TTL: 125

The three replies establish reachability during this sample. One request timed out before the replies began.

ARP resolution or routing convergence could explain an initial timeout, but the screenshot alone does not identify the cause. A repeated ping after the network settles is needed to assess steady-state loss.

Four responses, three forwarding routers

The destination is the fourth traceroute entry; this count is distinct from a RIP routing metric.



Observed response order

1. 7.0.0.254 - Router11
2. 4.0.0.2 - Router12
3. 5.0.0.2 - Router13
4. 6.0.0.1 - PC3

Hop	Responder	Probe 1	Probe 2	Probe 3
1	7.0.0.254	0 ms	0 ms	0 ms
2	4.0.0.2	0 ms	6 ms	0 ms
3	5.0.0.2	3 ms	0 ms	0 ms
4	6.0.0.1	0 ms	0 ms	0 ms

Command: `tracert 6.0.0.1`. Maximum shown: 30 hops. Result: Trace complete. A 0 ms display reflects simulator/timer resolution, not proof of zero latency.

Conclusions and verification checklist

What is established by the supplied evidence, and what should be captured next.

Supported conclusions

The two LANs communicate in the captured tests. The trace follows Router11 - Router12 - Router13. Router11 has a RIP v2 configuration with automatic summarization disabled. Ping success is 3 of 4 for this sample.

Limits of the evidence

The screenshots do not prove which routing protocol installed the forwarding entries, stable zero-loss operation, upper-branch failover or load sharing. Full configurations, routing tables and a repeat test would resolve these gaps.

Recommended checks before publishing the repository

```
show ip interface brief
show ip protocols
show ip route
show ip route rip
show running-config
```

On each router, verify interface state, RIP version, advertised networks and learned routes. On PC2, repeat ping and tracert to 6.0.0.1. Capture routing tables before and after a controlled link-down test to document alternate-path behavior.

RIP interpretation. RIP selects routes using hop count; its metric should not be read directly from the four traceroute entries. RIPv2 carries subnet information, and `no auto-summary` disables automatic classful summarization. [1, 2]

Sources and provenance

Primary evidence: the five supplied Packet Tracer screenshots, reproduced on pages 1, 4, 6 and 7. Page 2 adds the requested path overlay. No live simulator tests were run for this document.

[1] Cisco IOS IP Routing: RIP Command Reference [2] Cisco IP Routing: RIP Configuration Guide